

B.Sc. Semester-V
2024
Computer Science
CC-12: Theory of Computation

Time: 3 hours

Full Marks: 60

Answer question number 1 and any five from the rest

- 1 a) Distinguish between deterministic and nondeterministic finite automata.
- b) State Arden's theorem.
- c) Find the highest type number which can be applied to the following productions:

$$S \rightarrow Aa \quad A \rightarrow c \mid Ba \quad B \rightarrow b$$
- d) Find a regular expression corresponding to each of the following subsets of $\{a, b\}$.
 (i) The set of all strings containing exactly two a 's
 (ii) The set of all strings containing at most two a 's
- e) Distinguish between Mealy and Moore machines.

[2×5=10]

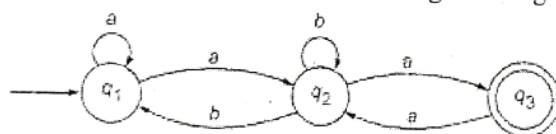
- 2 a) Construct a minimum state automaton equivalent to a DFA whose transition table is defined by the following table (q_0 is the starting state and q_3 and q_4 are the final states).

State	a	b
q_0	q_1	q_0
q_1	q_4	q_1
q_2	q_4	q_2
q_3	q_5	q_3
q_4	q_7	q_4
q_5	q_3	q_5
q_6	q_6	q_6
q_7	q_7	q_7

- b) Construct the finite automaton equivalent to the regular expression
 $(a + b)^*(ab + ba)(a + b)^*$

[5+5=10]

- 3 a) Construct a context-free grammar generating $L = \{a^m b^n \mid m > n; m, n \geq 1\}$
- b) Consider the following transition system. Find the strings it recognizes.



[5+5=10]

- 4 a) With suitable example, discuss comparison method to find equivalence of two finite automata.
- b) State Pumping lemma for regular sets. What is its application?

[5+5=10]

- 5 a) Let L be the set of all palindromes over $\{a, b\}$. Construct a grammar G generating L .
- b) Compare and contrast pushdown automata over finite automata.

[5+5=10]

- 6 a) Reduce the following grammar G to CNF. G is

$$S \rightarrow aAD, \quad A \rightarrow aB \mid bAB, \quad B \rightarrow b, \quad D \rightarrow d$$

- b) Design a Turing machine that concatenates two strings of 1 's.

[5+5=10]

7 a) Construct a DFA with reduced states equivalent to the regular expression

$$10 + (0 + 1 + 11^*)0^*1$$

[5+5=10]

b) Show that the set $L = \{a^{i^2} \mid i \geq 1\}$ is not regular.

8 Write short notes on the following:

a) Chomsky classification of languages

[5+5=10]

b) Elimination of unit productions
